LEARNER GUIDE

The perpendicular from the centre to a chord

Circle Theorems · Lesson 8. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Draw any **chord** AB across a circle and drop a **perpendicular** from the centre **O** to it. The foot of that perpendicular, M, always lands on the **midpoint** of the chord: the **perpendicular from the centre bisects the chord**, so AM = MB. The reason is the familiar one — the two radii OA and OB are equal, so the two right-angled triangles OMA and OMB are **congruent** (RHS), forcing their third sides to match. The **converse** holds too: a line from the centre to a chord's midpoint must be perpendicular to it. Once the perpendicular halves the chord you have a right-angled triangle whose hypotenuse is a radius, so **Pythagoras** links the radius, the distance from the centre, and half the chord. A companion fact: **equal chords are equidistant from the centre** (and vice versa).

What you are learning to notice

- The **perpendicular from the centre bisects the chord** (AM = MB).
- The **converse**: the line from the centre to a chord's midpoint is perpendicular to it.
- Radius, distance from the centre and half-chord form a **right-angled triangle** (Pythagoras).
- **Equal chords are equidistant** from the centre.

Moving through it, one step at a time

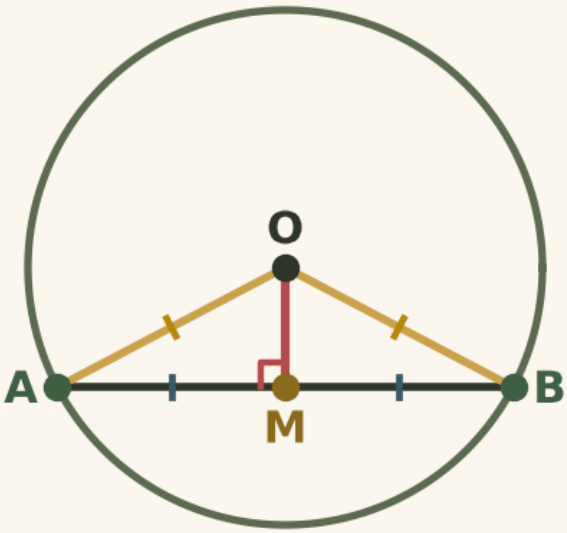
Spot a chord and the centre. Drop the perpendicular; it meets the chord at its midpoint, so mark equal halves and give the reason ('the perpendicular from the centre bisects the chord'). If you need a length, use the right-angled triangle: $\text{radius}^2 = (\text{distance from centre})^2 + (\text{half-chord})^2$. For two equal chords, their distances from the centre are equal.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Find a distance with Pythagoras

A chord AB of length 16 cm is drawn in a circle of radius 10 cm. Find the distance OM from the centre to the chord.

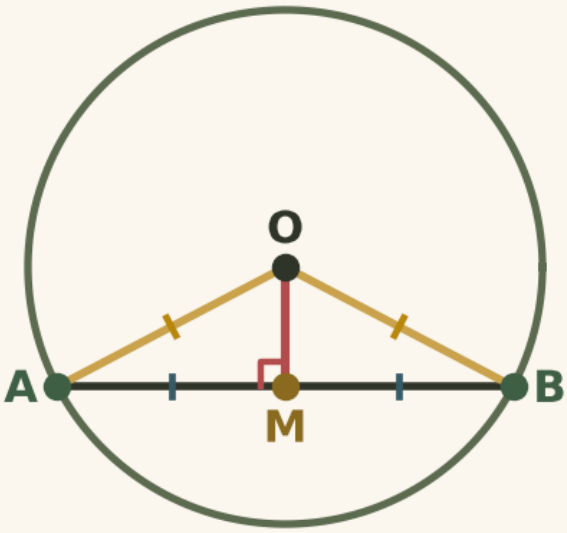


1. The perpendicular from O bisects AB, so AM = 16 ÷ 2 = 8 cm.	perpendicular bisects the chord
2. Triangle OMA is right-angled at M, with hypotenuse OA = 10 (radius).	radius is the hypotenuse
3. Pythagoras: $OM^2 = OA^2 - AM^2 = 10^2 - 8^2 = 36$.	$OM = \sqrt{36}$

OM = 6 cm — the chord is 6 cm from the centre.

Example 2 — Why the perpendicular bisects the chord

Prove that the perpendicular from the centre O to a chord AB bisects it (AM = MB).



1. Drop OM ■ AB, so ∠OMA = ∠OMB = 90°.	right angles at M
2. OA = OB (radii, tick marks) and OM is shared.	equal radii, shared side
3. Right angle, hypotenuse OA = OB, one shared side ⇒ congruent (RHS).	triangles OMA ≅ OMB

So AM = MB — the perpendicular from the centre bisects the chord. ■

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (that all radii of a circle are equal, congruent triangles (RHS), and Pythagoras' theorem), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.

- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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